

Manual of HDAC8 Assistant

open-source software

HDAC8 Assistant is a web application for the generation and evaluation of chemical compounds as potential HDAC8 inhibitors. The software combines de novo molecule generation (SMILES-RNN with reinforcement learning), QSAR-based activity prediction, applicability-domain assessment, drug-likeness evaluation (Muegge rules), and structural alert screening (Brenk filters and PAINS).

Appearance of the main window

The main page displays five capability panels: Molecule generation, Machine learning (QSAR), OECD guidelines (applicability domain), Muegge's rules, and Structural Alerts. A download button provides this manual.

Step 1. Draw molecule or select input molecular files

If you want to draw the structure of a chemical compound, use the built-in chemical editor Ketcher (<https://github.com/epam/ketcher>). After creating the structure, use two buttons: (1) "Reset" - delete the structure to create a new one; (2) "Apply" - transfer the structure for further analysis or generation in Step 2.

After clicking the "Apply" button, make sure that the structure has been created. If successful, its SMILES will be displayed under the chemical editor.

If you choose SMILES, paste the SMILES representation of the desired chemical structure and press Ctrl+Enter. If the entered structure is correct, the application will generate a 2D image of the compound. DON'T FORGET TO CLICK THE "APPLY" BUTTON when using the Ketcher editor.

If the entered structure is incorrect, the application reports an error. If Ketcher export fails aromatic bond assignment (kekulization), switch to the SMILES input method and paste the same structure as text.

If you choose a file *.sdf or *.csv containing multiple chemical structures, click the "Browse files" button and upload the file. For *.csv files, the file must contain a column named "SMILES".

If incorrect structures are detected in an *.sdf or *.csv file, the corresponding information appears in the section "CHEMICAL STRUCTURE VALIDATION AND STANDARDIZATION".

Step 2. Select analysis mode

In Step 2, choose one of the following modes from the drop-down menu:

- HDAC8 - predict HDAC8 inhibitor activity (pIC50) for the input structure(s).

- Molecu
structure fr

Step 2a. HDAC8 activity prediction

Select "HDAC8" and click the "Run predictions!" button. The QSAR model is a weighted CatBoost consensus of ECFP4, MACCS, PubChem and Klekota-Roth fingerprint families (late fusion; simplex weights). Applicability domain is nearest-neighbour Tanimoto distance on the concatenated binary fingerprints, cutoff = mean + 0.5*SD of training nearest neighbours.

Step 3. Prediction results (HDAC8 mode)

The form of presentation depends on the input format. For a single molecule (Ketcher or SMILES), results are displayed on screen. For *.sdf or *.csv files, results for valid structures are shown in a downloadable table.

The final table "Prediction results" contains the following columns:

- SMILES - chemical structure in SMILES notation.

- Pred
concentrati

Step 2b. Molecule generation (SMILES-RNN)

Select "Molecule generation (SMILES-RNN)" in Step 2. This mode generates new candidate HDAC8 inhibitors using a SMILES-RNN model pretrained on drug-like compounds from ChEMBL. The model is optimized via reinforcement learning (REINVENT algorithm) to maximize predicted pIC50 values while ensuring applicability-domain compliance. A weighted CatBoost consensus QSAR model serves as the core of the RL reward function.

Input requirements:

- Single reference: draw a molecule in Ketcher or enter one SMILES in Step 1.
 - Multiple job.
- Maximum molecules to return (default 500) - per reference, the maximum number of AD-compliant candidates kept from each RL run.
 - RL n_st

Step 3. Generation results

Generated structures undergo deduplication, validity checks, and optional ring-system filtering. Only molecules inside the QSAR applicability domain ($\text{in_AD} \geq 0.5$) are retained. Compounds matching structures in the experimental HDAC8 dataset (HDAC8_exp_data_inchi.csv) are excluded. Results are ranked by predicted pIC50 (descending).

The results table contains the following columns:

- No. - row number.

- generat

For multi-reference CSV input, results from all references are merged, deduplicated by canonical SMILES, sorted by predicted pIC50, and truncated to the chosen top N rows.

Step 2c. Muegge rules and structural alerts

Select "Muegge rules, PAINS, Brenk structural alerts, Substructural search" and click "Run predictions!". HDAC8 Assistant evaluates compliance with Muegge's drug-likeness rules. Results are displayed in tabular form and as a Bioavailability Radar for visual assessment of lipophilicity (log P), molecular weight (MW), hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), rotatable bonds, TPSA, and ring count against Muegge limits.

The application also detects widely used medicinal-chemistry structural alerts:

- Brenk filters - 105 fragments identified by Brenk et al. [Brenk, R. et al. Lessons learnt from assembling screening libraries for drug discovery for neglected diseases. ChemMedChem 3, 435-444 (2008).] as putatively toxic, chemically reactive, metabolically unstable, or associated with poor pharmacokinetics.

- PAINS
Holloway,
screening
false-positi

Software requirements for molecule generation

Molecule generation requires the Python packages smiles-rnn, molscore, and PyTorch (see requirements.txt). Optional packages useful-rdkit-utils (ring-system filtering) and sascorer (SAScore calculation) enhance post-processing when available.